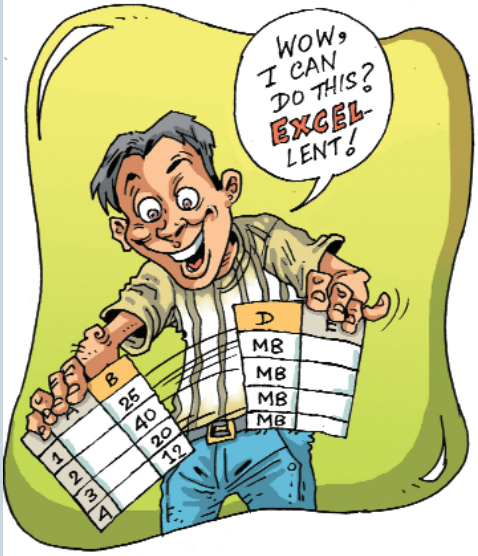
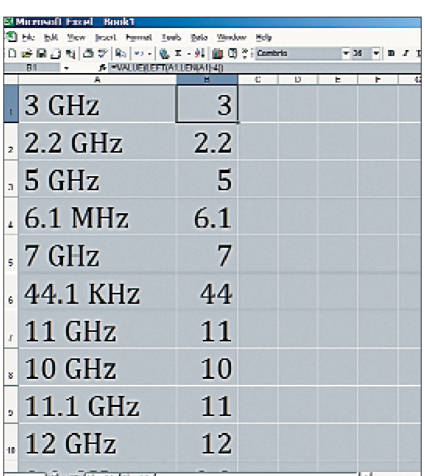
FORMULAS IN EXCEL



Using The Numbers #1

Think of when you have values such as "3.2 GHz" in numerous cells. You obviously can't use that "3.2" as it is—you'll need to "extract" it from the full string. There happens to be a way to do this. Assume the entries all consist of a number followed by a space, followed by some characters. You can perform the extraction by following these steps: first, make sure there's a blank column to the right of the entries. Then select the entries. Choose "Text to Columns" from the Data menu. Excel will display the first step of the Convert Text to Columns Wizard.

Select the "Delimited" radio button and click Next. Excel will display the second step of the wizard. Now, making sure the Space checkbox is selected, click Next. The third step of the wizard will be displayed; click Finish.



	A	B
1	3 GHz	3
2	2.2 GHz	2.2
3	5 GHz	5
4	6.1 MHz	6.1
5	7 GHz	7
6	44.1 KHz	44
7	11 GHz	11
8	10 GHz	10
9	11.1 GHz	11
10	12 GHz	12

Separating numbers and text, so that you can use the numbers for calculations

Excel splits the entries into two columns, with the numbers in the left one and the alphabetic characters in the right. You can then run any functions on the numeric values!

Using The Numbers #2

If it doesn't seem a good idea to separate the data into columns—for example, if it might cause problems with others using the worksheet—then a different approach is needed. There are two ways you can do the extraction mentioned in the previous tip.

You can use this formula on individual cells:

=VALUE(LEFT(A1,LEN(A1)-4))

The LEFT function strips off the three rightmost characters (the space and the "GHz") of whatever is in cell A1, and then the VALUE function converts the result to a number. You can then use this result as you would use any numeric value.

Using The Numbers #3

On the same problem, if all you want to do is sum the column containing your entries, you could use an array formula. Enter the following formula into a cell:

=SUM(VALUE(LEFT(A1:A10,LEN(A1:A10)-4)))

You need to make sure you actually enter the formula by using [**Shift**] + [**Ctrl**] + [**Enter**].

Using Names

Excel allows you to define names that can refer to formulas or constants. This can be useful in several cases: suppose you use a constant, say an interest rate equal to 12 per cent, quite often in your worksheet. To define a name for that 12 per cent, you'll need to do the following: first, select the Name option from the Insert menu and choose Define from the sub-menu. Excel will display the Define Name dialog box. Change the Refers To field at the bottom of the dialog box so that it contains the desired formula—that is, "=12%". Click on Add, and your name will have been defined. Click OK.

You can now use the constant in formulas. For example, if you used "digit" in the Name field, you can now say "=(50*digit)" in a cell, and you'll get 6.25 in that cell.

Using An Input Mask...

When you're keying in times into cells using the numeric keypad, what slows you down is the need to enter the colons. There happens to be a way to use an input mask so you can enter values such as 6:30 by typing in just "630"! Use the following procedure:

First select the cells you want to use for inputting the time. Then go to **Format** > **Cells**. Under the Number tab, in the Category list, choose Custom. Replace whatever is in the Type box with **#":"00** (a hash sign, a quote mark, a colon, another quote mark, and two zeroes). Click OK. You can now enter your times using only digits!

...And Then A Formula

The problem with the above tip, of course, is that the cells don't really contain times. If you entered "630" for 6:30, it doesn't contain 6:30 as a time—it contains *six hundred and thirty*. As a result, you can't use the contents of the cell in time calculations.

In order to get around this, you can use a separate column to show the entered numbers converted into a time. You'll need a formula for this. Say the time you entered is in cell B1; use the following:

=(INT(B1/100)/24)+((B1—(INT(B1/100)*100))/1440)

Now you just need to format the cell that contains this formula so it displays a chosen time format.

[Alt] + [Enter] In A Formula

When you're entering data in a cell, you can use [**Alt**] + [**Enter**] to start a new line within the current cell. You may want to create a new line in a formula as well, just as if you had pressed [**Alt**] + [**Enter**].

Say you're working with the formula that concatenates text values: **=A1 & A2 & A3**, and you need a way to make it appear on separate lines. You can use the actual character code that Excel uses when you press [**Alt**] + [**Enter**], as in the following formula:

=A1 & CHAR(10) & A2 & CHAR(10) & A3

The "**CHAR(10)**" inserts a line feed character. If you don't see the results on separate lines, it's because you haven't turned on wrapping for the cell. You'll be seeing a square box where the line feed character is located. To see the results of the formula on separate lines, go to **Format** > **Cells** > **Alignment** and check the Wrap Text box.

Numbers With Signs #1

As you work with a large set of numbers consisting of both negative